

**Paper Practice 2.2 • WORKED KEY**  
**Solving One-Step Equations**

**WORKED KEY (odd-numbered items) – Undo the operation on the variable with its opposite, on BOTH sides. Check by substituting your answer back in. Even-numbered items are yours to work.**

**Solve each equation.**

1.  $36 + x = 63$

**Answer:**  $x = 63 - 36 = 27$

2.  $-5d = 65$

3.  $43 - (-c) = 32$

**Answer:**  $43 + c = 32 \rightarrow c = 32 - 43 = -11$

4.  $\frac{m}{15} = -6$

5.  $h - 12 = 5$

**Answer:**  $h = 5 + 12 = 17$

6.  $7k = 42$

7.  $\frac{2}{5}y = -10$

**Answer:** multiply both sides by  $\frac{5}{2}$ :  $y = -10 \cdot \frac{5}{2} = -25$

8.  $33 = g + (-15)$

9.  $110 = 11w$

**Answer:**  $w = 110 \div 11 = 10$

10.  $13 = -22 + y$

11.  $-16 = -\frac{b}{2}$

**Answer:** multiply both sides by  $-2$ :  $b = 32$

12.  $t - (-4) = 13$

13.  $\frac{5}{6} = p - \frac{4}{3}$

**Answer:**  $p = \frac{5}{6} + \frac{4}{3} = \frac{5}{6} + \frac{8}{6} = \frac{13}{6} = 2\frac{1}{6}$

14.  $9 = -\frac{3}{2}z$

15.  $p + (-6) = -4$

**Answer:**  $p = -4 + 6 = 2$

16.  $\frac{a}{7} = -\frac{5}{14}$

17.  $14 = 26 + t$

**Answer:**  $t = 14 - 26 = -12$

18.  $9 = -\frac{1}{5}q$

19. **FINANCES** Margo is repaying a loan from her parents. She has already paid \$315 and still owes them \$1245. Write and solve an equation to find the original amount  $a$  of the loan.

**Answer:**  $a - 315 = 1245 \rightarrow a = 1245 + 315 = \$1560$

20. **DISCOUNTS** Gustavo has been waiting to purchase an acoustic guitar. This week, the guitar is on sale for  $\frac{3}{4}$  of the original price. With the sale, Gustavo paid \$90, without tax, for the guitar. Write and solve an equation to find the original price  $p$  of the guitar.